BEING SEPARATED IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one lemma from Morphisms of Spaces, one from Topologies on Spaces, one same-chapter lemma and one definition, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Base change of separated morphisms. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-separated. Supplies the direct implication.
- fpqc coverings are stable under base change. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-fpqc. Supplies the covering of X \times_Y X.
- fpqc descent of the closed immersion property (diagonal form). Stacks Project, Descent on Spaces, Lemma lemma-descending-property-closed-immersion. Supplies the descent step from the \Delta_i to \Delta.

Source reference. Original reference 03KL: sources/stacks-chow/tags.txt, source lines 4681–4681. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03YH: sources/stacks-chow/tags.txt, source lines 5184–5184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0420: sources/stacks-chow/tags.txt, source lines 5311–5311. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is separated" is fpqc local on the base.*

Proof. A base change of a separated morphism is separated, see Morphisms of Spaces, Lemma 03KL. Hence the direct implication in Definition 03YH.

Let $\{Y_i \to Y\}_{i \in I}$ be an fpqc covering of algebraic spaces over S. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume each base change $X_i := Y_i \times_Y X \to Y_i$ is separated. This means that each of the morphisms

$$\Delta_i : X_i \longrightarrow X_i \times_{Y_i} X_i = Y_i \times_Y (X \times_Y X)$$

is a closed immersion. The base change of a fpqc covering is an fpqc covering, see Topologies on Spaces, Lemma 03MR hence $\{Y_i \times_Y (X \times_Y X) \to X \times_Y X\}$ is an fpqc covering of algebraic spaces. Moreover, each Δ_i is the base change of the

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morphism $\Delta : X \to X \times_Y X$. Hence it follows from Lemma 0420 that Δ is a closed immersion, i.e., f is separated. $\square$